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ROLL NO .33 SOC-15

Q1.Write a program to make use of basic I/O functions using different data types.

Ans.

```
#include <stdio.h>

int main()
{
    int age;
    float marks;
    char grade;
    double salary;

    printf("Enter your age: ");
    scanf("%d", &age);

    printf("Enter your marks: ");
    scanf("%f", &marks);

    printf("Enter your grade: ");
    scanf(" %c", &grade);

    printf("Enter your salary: ");
    scanf("%lf", &salary);

    printf("\n--- Student Information ---\n");
    printf("Age = %d\n", age);
    printf("Marks = %.2f\n", marks);
    printf("Grade = %c\n", grade);
    printf("Salary = %.2lf\n", salary);

    return 0;
}
```

OUTPUT-

```
Enter your age: 18
Enter your marks: 85.5
Enter your grade: A
Enter your salary: 25000

--- Student Information ---
Age = 18
```

Marks = 85.50
Grade = A
Salary = 25000.00

Q2.Write a program to calculate the area of a circle.

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```
#include <stdio.h>

int main()
{
   float radius, area;
   const float PI = 3.14;

   printf("Enter the radius of the circle: ");
   scanf("%f", &radius);

   area = PI * radius * radius;

   printf("Area of the circle = %.2f\n", area);

   return 0;
}
```

OUTPUT-

Enter the radius of the circle: 5
Area of the circle = 78.50

Q3.Write a program to calculate the average of three numbers.

```
#include <stdio.h>

int main()
{
    float a, b, c, average;

    printf("Enter three numbers: ");
    scanf("%f %f %f", &a, &b, &c);

    average = (a + b + c) / 3;

    printf("Average = %.2f\n", average);

    return 0;
}
```

OUTPUT-

```
Enter three numbers: 10 20 30
Average = 20.00
```

Q4.Write a C program to swap two numbers using a temporary variable.

```
#include <stdio.h>

int main()
{
    int a, b, temp;

    printf("Enter two numbers: ");
    scanf("%d %d", &a, &b);

    printf("Before swapping: a = %d, b = %d\n", a, b);

    temp = a;
    a = b;
    b = temp;

    printf("After swapping: a = %d, b = %d\n", a, b);

    return 0;
}
```

OUTPUT-

```
Enter two numbers: 10 20
Before swapping: a = 10, b = 20
After swapping: a = 20, b = 10
```

Q5.Write a C program to demonstrate the use of single-line and multi-line comments.

```
#include <stdio.h>

int main()
{


    printf("Hello, World!\n");
    printf("Demonstration of comments in C.");

    return 0;
}
```

OUTPUT-

```
Hello, World!
Demonstration of comments in C.
```